

have only your office computer on which you can save the pictures taken that day. Since you can't keep them in the main photo collection, you put them inside the C:\Windows\System32\Config folder. That's clever since no one will ever look for pictures in that folder. Did you just do some crypto? Well, the answer is no.

Cryptography is the science of scrambling information. Hiding information cannot be called as cryptography. Hence, uploading files to your Dropbox or Google Drive account whose password is unknown to anyone but you would still not qualify as cryptography.

It does, however, if you zipped the set of pictures into a zip file with a password. This is because the software you used for zipping with a password (e.g. Windows shell or WinZip) actually uses an algorithm called AES to encrypt the data – in other words, it modifies the data.

You might ask “Even zipping pictures into a zip file without a password would modify the data. Why doesn't that count as cryptography?” Remember that cryptography doesn't just involve ‘altering/modifying the data’. It is a method to ‘alter the data in such a way that unintended people are unable to interpret it’. Given the easy availability of compression software which can easily detect a ZIP file (even if you change the extension), just compressing the picture set won't protect it.

Another question might arise at this point. At the beginning of this chapter, we used a technique of rotating the letter of the alphabet – is that a part of cryptography given that it was even easier to crack than it would be by compressing/decompressing it? Well, it depends. Altering information to render it incomprehensible is one part of cryptography. If the trial and error method to understand the message works and you didn't need to know that rotating the letters of the alphabet was the key, then it's no cryptography at all. But to someone who can't figure out what those random letters mean, the message remains unbreakable and thus, successful confidentiality is achieved. Once again though it depends, on who's attempting to break the code. Given the intelligence of our species which aims to colonize Mars, it wouldn't be an unimaginable feat for people of a certain level of intelligence. In such a case, the message encrypted using the ‘alphabet rotation’ technique is vulnerable to being intercepted and would qualify as a weak cryptographic method.

Breaking down the jargon

If you've been using computers since some time and have read this far, you might wonder who really uses these techniques in real life. You're right,